

Computer Networks
CL3001

LAB - 05
SMTP & FTP
Wireshark

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES,
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Computer Networks Lab 05

Course: Computer Networks (CL3001)
Instructor: Sadaf Zehra

Semester: Fall 2026
T.A: N/A

Note:

- Maintain discipline during the lab.
 - Listen and follow the instructions as they are given.
 - Just raise hand if you have any problem.
 - Completing all tasks of each lab is compulsory.
 - Get your lab checked at the end of the session.
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Lab Objective

- Introduction to SMTP & FTP in Cisco Packet Tracer.
- Network traffic analysis of HTTP/HTTPS protocol headers, cookies using Wireshark.

Lab Exercise SMTP and FTP:

1. Let's suppose your organization needs to create its small server (to provide some services) based network. With below mentioned topology and instructions:

Attach a screenshot of each step.

Also, Submit a .pkt file.

- a) Configure **SMTP** (create an account with your last name along with the last 3 digits roll number) and send mail from PC A to PC B and PCB to PC A.
- b) Add 4 PC's in the network and then Send email between any of two.
- c) Configure the **FTP** server to create an account with your first name, password with your roll number, and filename with your last name (.bin extension) to show all connection results. The **FTP** Server should be established on both Server 1 and Server 3.

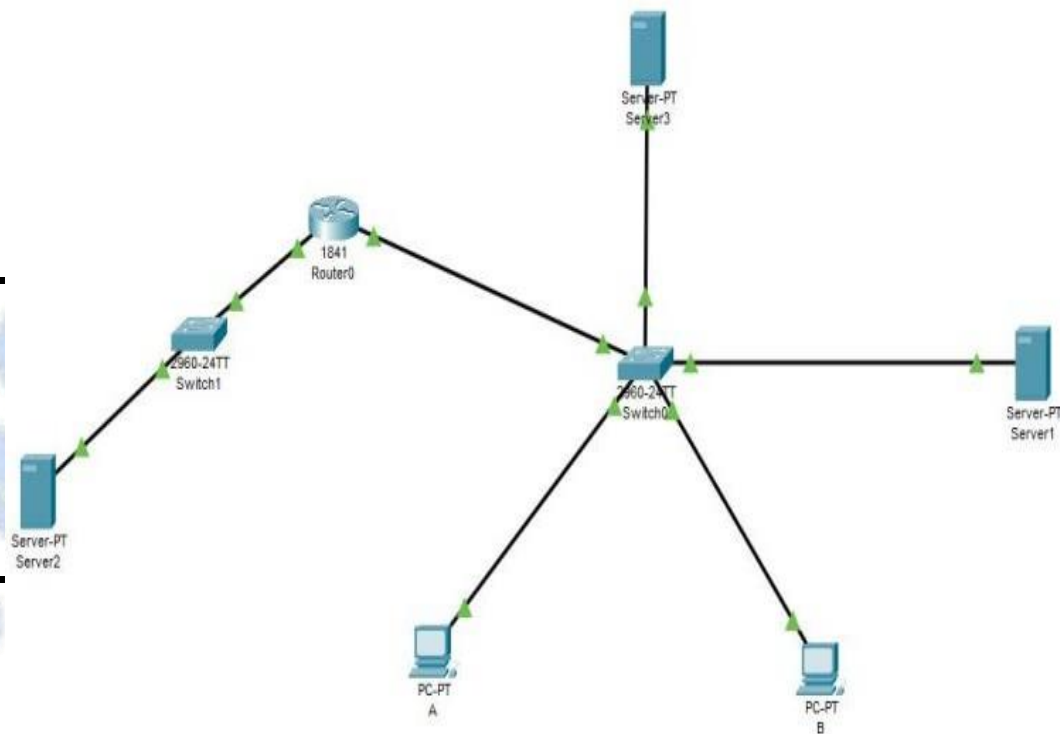


Figure 1

Wireshark Lab Exercise

Task 1: DNS Analysis

Capture DNS traffic in Wireshark and analyze the **DNS request and response packets**. Apply the `dns` filter.

Task 2: IP Address Filter

Capture network traffic in Wireshark and apply an **IP address filter** to display packets related to a specific IP address.

Task 3: Destination IP Filter

Apply a **destination IP filter** in Wireshark to identify packets being sent to a specific destination IP address.

Task 4: Protocol Filter

Apply a **protocol filter** such as `http`, `tcp`, or `dns` in Wireshark.

Task 5: Source and Destination IP Analysis

Apply a filter using both **source and destination IP addresses** to identify communication between two devices.

Task 6: Packet Details Analysis

Select any captured packet and inspect its **Packet Details** and **Packet Bytes** sections.